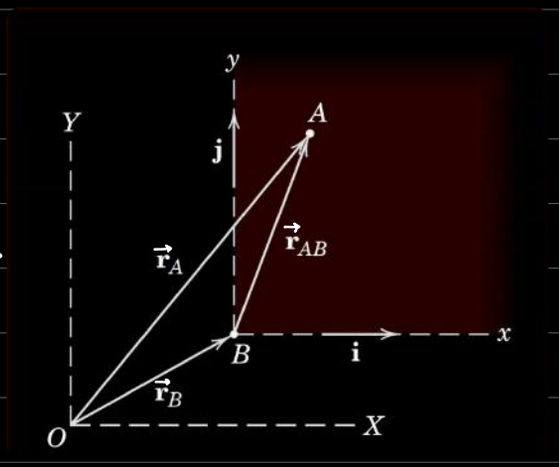
RELATIVE POSITION: We introduced the idea of position vectors and relative position in our study of vectors. Here let us recall the definition. Let us say there are two particles A and B having position vectors $\vec{r}_A$ and $\vec{r}_B$ then we define position vector of A relative to B as a vector from B to A. This is represented by symbol $\vec{r}_{AB}$. See the figure above. Using triangle law of addition we can write that $\vec{r}_B + \vec{r}_{AB} = \vec{r}_A$ thus,



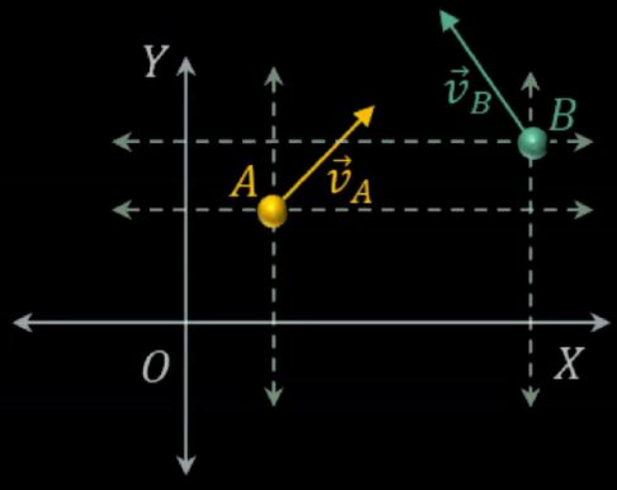
$$\boxed{\vec{r}_{AB} = \vec{r}_A - \vec{r}_B}$$

① Similarly $\vec{r}_{BA} = \vec{r}_B - \vec{r}_A$.

$\vec{r}_{AB}$ is also read as position of A as seen by B because that's precisely what it physically represents. A is located at a distance r_{AB} from B in the direction $\hat{r}_{AB}$.

RELATIVE VELOCITY:

Rate of change of relative position wrt time is known as relative velocity. Mathematically, $\vec{v}_{AB} = \frac{d}{dt}(\vec{r}_{AB})$.



Now $\vec{r}_{AB} = \vec{r}_A - \vec{r}_B$, Differentiating wrt time on both sides:

$\dot{\vec{r}}_{AB} = \dot{\vec{r}}_A - \dot{\vec{r}}_B$, thus:

$$\boxed{\vec{v}_{AB} = \vec{v}_A - \vec{v}_B}$$

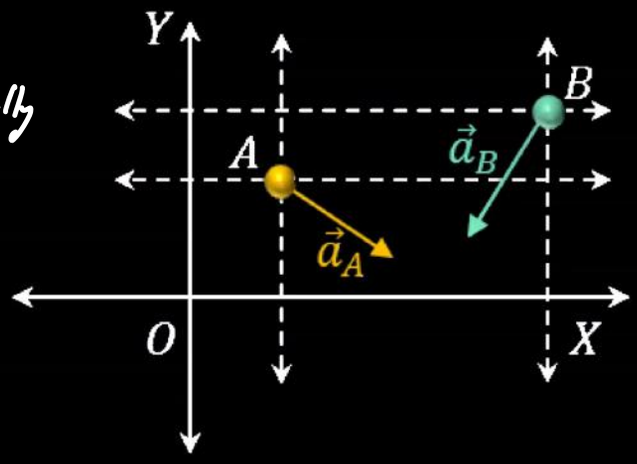
Physically it represents velocity of A as visible from frame of B.

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Relative Acceleration:

Rate of change of relative velocity is called relative acceleration. Mathematically $\vec{a}_{AB} = \dot{\vec{v}}_{AB} = \dot{\vec{v}}_A - \dot{\vec{v}}_B$ thus:

$$\boxed{\vec{a}_{AB} = \vec{a}_A - \vec{a}_B}$$



physically it represents the acceleration of A as visible from B.

Condition For Collision Between Uniformly Accelerated Particles:

For collision between particles, their position vectors must become same at some point of time. Let A and B be two particles under observation then:

$\vec{r}_A = \vec{r}_{0A} + \vec{u}_A t + \frac{1}{2}\vec{a}_A t^2$ similarly,

$\vec{r}_B = \vec{r}_{0B} + \vec{u}_B t + \frac{1}{2}\vec{a}_B t^2$, Now for collision $\vec{r}_A - \vec{r}_B = 0$

$\Rightarrow (\vec{r}_{0A} - \vec{r}_{0B}) + (\vec{u}_A - \vec{u}_B)t + \frac{1}{2}(\vec{a}_A - \vec{a}_B)t^2 = 0$ thus:

$$\boxed{\vec{r}_{0AB} + \vec{u}_{AB}t + \frac{1}{2}\vec{a}_{AB}t^2 = 0}$$

→ For projectile collision problems, if both are simultaneously in air at $t=0$, this equation gets simplified since $\vec{a}_{AB} = 0$.

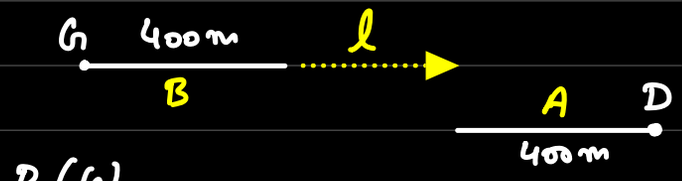
→ One dimensional overtaking problems can be modelled as collision problems (essentially for overtaking also we are making coordinate of tail of one vehicle coincide with coordinate of head of another vehicle, though vehicles are on parallel lines, so don't actually collide).

→ For easy visualization of problems, we sometimes introduce hypothetical particles whose frame may make the observed particle's initial velocity or acceleration zero.

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EXAMPLE:

Two trains A and B of length 400 m each are moving on two parallel tracks with a uniform speed of 72 km/h in the same direction, with A ahead of B. The driver of B decides to overtake A and accelerates by 1 m/s². If after 50 s, the guard of B just brushes past the driver of A, what was the original distance between the guard of B & driver of A?



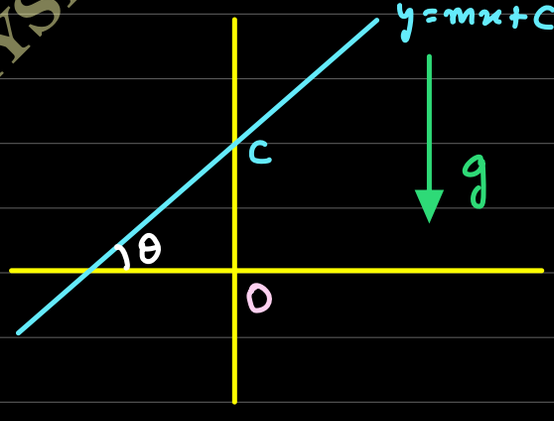
Solution: Let the trains be separated by an initial distance l. The problem involves making coordinates of Guard of B (G) same as coordinate of driver of A (D). Thus:

$\vec{s}_{0DG} + \vec{u}_{DG}\,t + \frac{1}{2}\vec{a}_{DG}\,t^2 = 0$, we chose rightward direction as positive, thus:

$$(800+l) + 0 \times 50 + \frac{1}{2} \times (-1) \times 50^2 = 0 \quad (1)$$

($\because a_{DA} = a_D - a_A = -1$). Solving (1), we get $800 + l = 1250\,m$ Ans.

EXAMPLE: A projectile is to be thrown from the origin so as to hit the line $y = mx + C$ in the shortest possible time. (a) In which direction should the particle be launched (b) (original), what is the minimum required speed for hitting the line, and the corresponding time of hit.



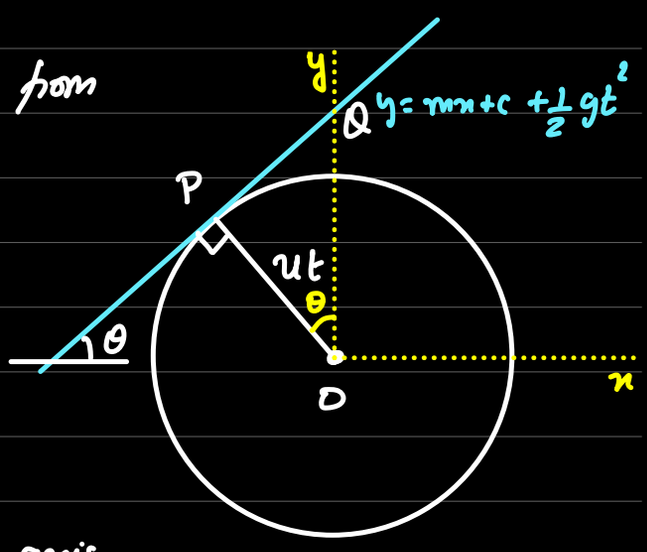
Solution: To make the visualisation simpler, imagine a frame falling with acceleration g downwards with initial velocity zero. In such a frame, our projectile will appear to move with constant velocity (in a straight line), and the line will appear to move upwards with an acceleration g. Let the projection speed of the particle be u. Now depending on the launching direction, at a general time t, the particle can be anywhere in a circle of radius ut, and for hitting, the line should just start touching this circle, i.e it should be tangent to the circle. If the origin of our falling frame is made to coincide with initial position

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of our projectile, then the collision from this frame is visualised as shown in adjacent figure. Thus direction of launch required from this frame is along ⊥ to the line i.e at an angle $90° + \theta$ with x axis where $\tan\theta = m$.

Thus required direction of launch:

$\phi = 90° + \tan^{-1} m$ from x axis.



Now time for hitting: $t = OP/u = OQ\cos\theta/u$ i.e

$t = (c + \frac{1}{2}gt^2)\cos\theta/u$, rearranging we get:

$\frac{1}{2}gt^2\cos\theta - ut + c\cos\theta = 0$, the time for hitting must be a real number, therefore, the discriminant of this equation must be positive, thus:

$$u^2 - 2gc\cos^2\theta > 0 \Rightarrow u^2 > 2gc\cos^2\theta$$

i.e $u^2 > \frac{2gc}{1+\tan^2\theta} \Rightarrow u > \left(\frac{2gc}{1+m^2}\right)^{1/2}$ Ans.

If u is exactly equal to this, then time for hitting:

$T = u/g\cos\theta$ i.e ($\because -b/2a$).

$$T = \left[\frac{2c(1+\tan^2\theta)}{g(1+m^2)}\right]^{1/2} = \left[\frac{2c}{g}\right]^{1/2}$$

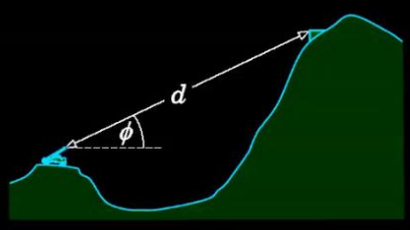
→ This is the maximum time in which it is possible to hit the line.

→ If we choose any other direction of launch, the time for hitting becomes longer than the time for hitting along perpendicular. Thus if perpendicular hitting time becomes imaginary with a given u, then launchings in any other direction will also yield an imaginary time. (One can also see from the falling frame that component of velocity parallel to the line is useless for hitting).

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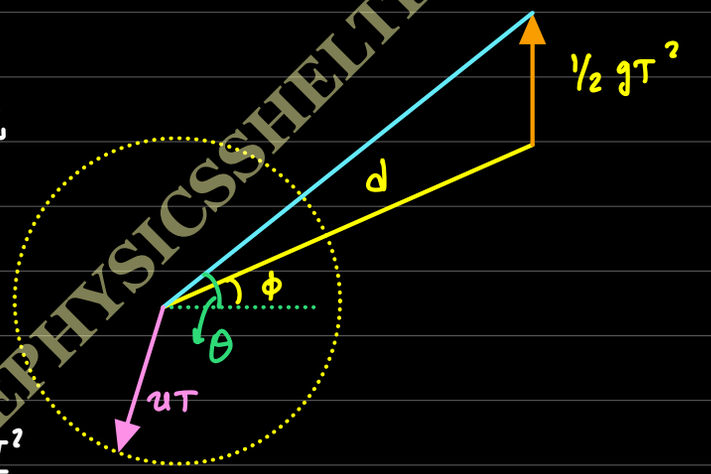
Example: Pathfinder: Build 19:

A cannon installed at the top of a hill can fire shells in all directions. There is an enemy bunker at an angle of elevation ϕ and a distance d from the cannon. All the shells fired explode in air in time T before they reach the bunker. At what angle to the horizontal, should a shell be fired with a speed u to explode closest to the bunker? Acceleration due to gravity is g.



Solution: Here again think of a frame falling freely with an acceleration g downwards starting with an initial velocity zero. In such a frame the shot could be anywhere on periphery of a circle of radius uT where u is the launch velocity while the bunker will have moved upwards a distance $\frac{1}{2}gT^2$.

This is graphically shown in adjacent figure. Now we just need to launch the shell towards a point on circle which is nearest to the bunker's position at time T. Thus, shell needs to be launched at an angle θ such that, $\tan\theta = \frac{d\sin\phi + \frac{1}{2}gT^2}{d\cos\phi}$ Ans.

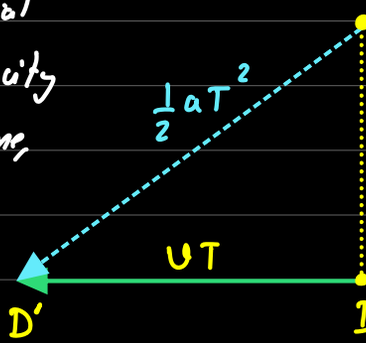


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Example: Pathfinder: Check-28:

A honeybee is flying parallel to a tabletop at a height $h = 2.0$ m with a constant velocity $v = 20\sqrt{2}$ m/s. With its wings, it can achieve a maximum acceleration $a = 400\sqrt{3}$ m/s^2. At an instant when the honeybee is vertically above a honey drop on the tabletop, it decides to reach the honey drop. Neglect the reaction time of the honeybee and find the minimum time in which the honeybee can reach the honey drop.

Solution: Let us consider a frame coinciding with initial position of bee and moving with a constant velocity equal to initial velocity of bee. In such a frame, the drop D appears to be moving backwards with a speed v and the bee has to start from rest. The fastest chase is then along straight line BD'. Now from the figure,



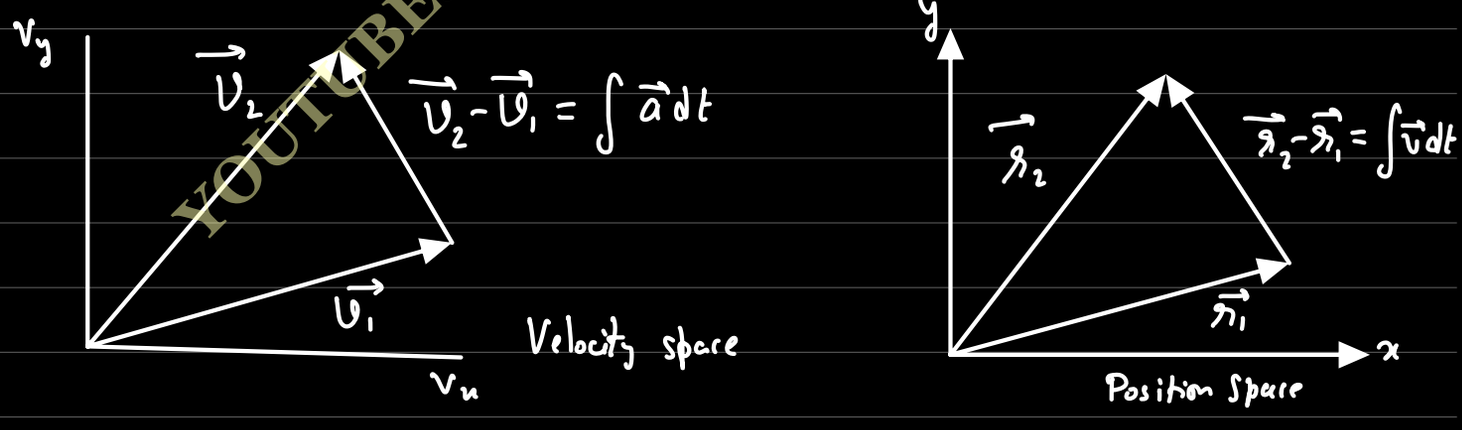
$$\frac{1}{2}aT^2 = \sqrt{h^2 + (vT)^2} \quad \Rightarrow \quad T = \sqrt{\frac{2(v^2 + \sqrt{v^4 + h^2a^2}}{a}} = 0.1 \text{ sec Ans.}$$

→ Think: What will path of bee look like from ground frame?

Ans: Just like that of a projectile with initial velocity horizontal an inclined acceleration.

EXAMPLE: (ORIGINAL): A car is moving with a speed u towards east. It wants to change it velocity towards 30° North of east in shortest possible time. If magnitude of its acceleration cannot exceed a, find the least possible time to achieve the desired objective.

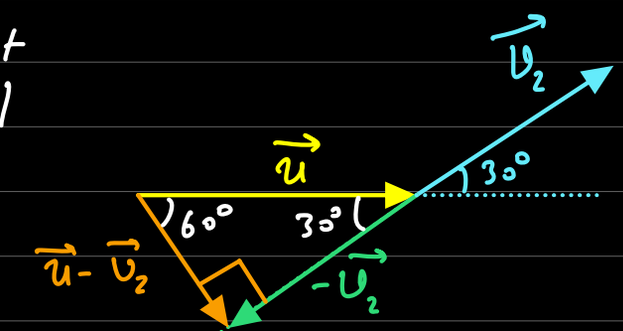
Solution: Let us see the problem from frame of initial velocity of the car (say $\vec{v}_1$), from this frame the car is initially at rest, and if its final velocity is $\vec{v}_2$ from ground frame, then from our frame its final velocity is $\vec{v}_2 - \vec{v}_1$. Thus problem reduces to achieving a velocity $\vec{v}_2 - \vec{v}_1$ from rest. Just as to achieve a displacement in least possible time with an upper bound on speed, all the time we must move in the direction of desired displacement, in the same manner, to achieve a desired velocity change, we must accelerate along the total change in velocity all the time.



This can be easily understood from analogous figures in velocity space and position space. Thus least required time to achieve $\vec{v}_2$ starting from $\vec{v}_1$ is simply $|\vec{v}_2 - \vec{v}_1|/a$. Now in our case $|\vec{v}_2|$ is not fixed although direction of $\vec{v}_2$ is fixed. Thus to achieve the objective in least possible time, we need to minimize

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$|\vec{v}_2 - \vec{v}_1|$. Now as clear from adjacent vector diagram, least value of $|\vec{u} - \vec{v}_2|$ occurs when $\vec{u} - \vec{v}_2$ is chosen $\perp$ to $\vec{u}$. Thus $(\vec{v}_2 - \vec{u})_{min} = u \sin 30° = u/2$.



Thus required minimum time is $T = u/2a$ Ans.

The direction of this acceleration must be in direction of $\vec{v}_2 - \vec{u}$ i.e 60° north of west. (We wanted to minimize $|\vec{v}_2 - \vec{u}|$ which is effectively same as minimizing $|\vec{u} - \vec{v}_2|$.

→ Since it is a motion with constant accelerations, the path will be parabolic.

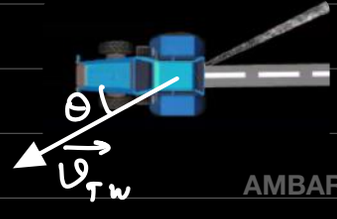
EXAMPLE:



A tractor is moving with a constant speed along a road with constant speed such that its smoke trail makes an angle θ with road as shown in figure. A horizontal wind is blowing with a constant speed u in a direction $\perp$ to the smoke trail. What is the speed of the tractor?

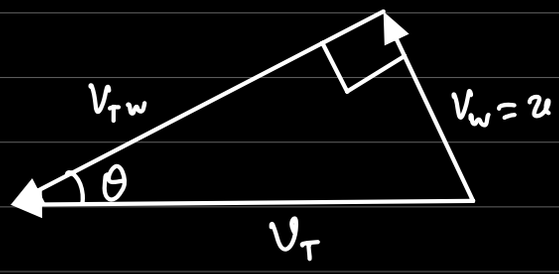
Solution: From the wind frame, smoke trail appears stationary and the tractor appears to move with some velocity collinear with smoke trail, (say $\vec{v}_{TW}$)

Now $\vec{v}_{TW} = \vec{v}_T - \vec{v}_W$ W: WIND, T: TRACTOR



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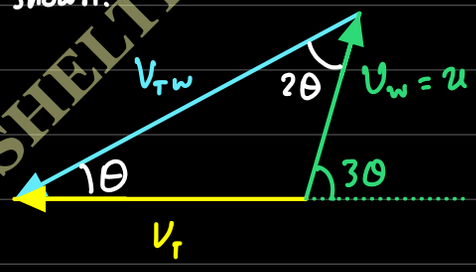
Thus, $V_T = \vec{V}_w + \vec{V}_{Tw}$, Now drawing the vector triangle:



Thus, $V_T = W_w \cos\theta = u\cos\theta$ Ans.

EXAMPLE: Repeat the previous problem if the wind were blowing at angle 3θ North of east.

Solution: Now our triangle diagram is modified as shown.



Now applying sine law:

$$\frac{V_T}{\sin 2\theta} = \frac{u}{\sin\theta} \Rightarrow V_T = \frac{u\sin 2\theta}{\sin\theta} = 2u\cos\theta \quad \text{Ans.}$$

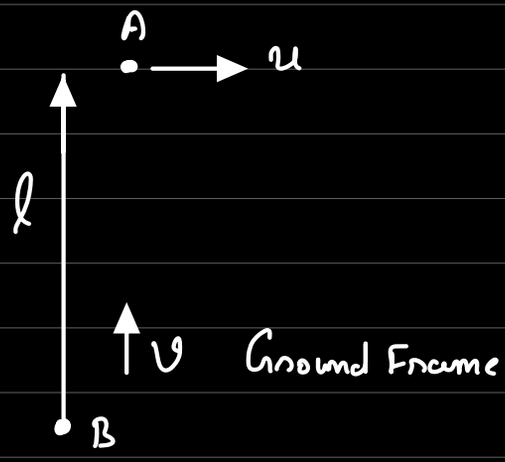
Think: If a flag were mounted on the tractor, in which direction will it flutter?

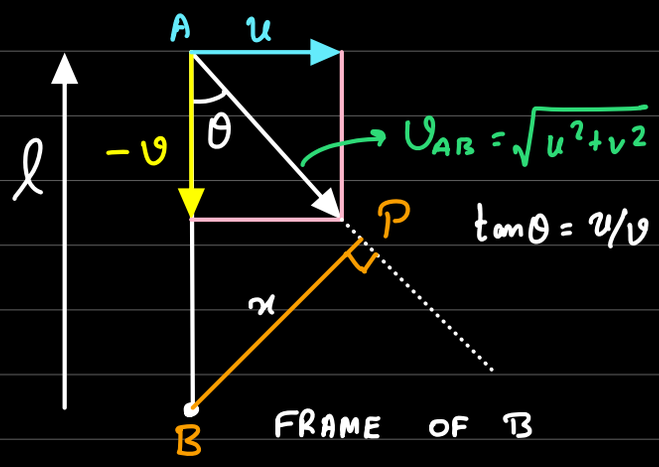
Ans: Same direction as smoke trail. Smoke trail is in direction of wind as seen from tractor frame & similarly flag flutters in direction of wind as seen from the tractor frame.

EXAMPLE: Two ships A and B are ℓ distance apart on a north south line. Ship A which is north of B is steaming East with a speed u while ship B is steaming northwards with a speed v. Assuming speeds to be constant, (a) What is their closest distance of approach, (b) After how much time is their distance minimum?

Solution: From frame of B, ship B appears stationary while A appears to move with a constant velocity $\vec{V}_A - \vec{V}_B$ along a straight line. Let us visualize this by drawing a figure.

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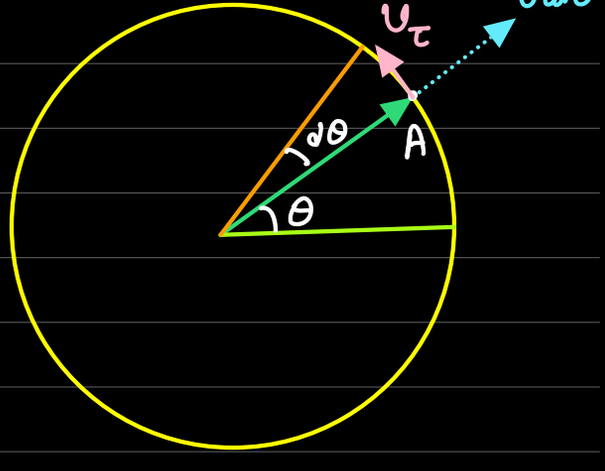
Refer to the vector diagram in the figure. From frame of B, A appears to move towards south east such that $\tan\theta = u/v$. Now nearest distance is found by drawing ⊥ on line of motion of A as seen from B. Thus nearest distance, $x = BP = l\sin\theta$

Thus min distance $= x = \dfrac{u}{\sqrt{u^2+v^2}} \times l$ Ans.

Now to reach this point, ship A has to cover a distance AP with a speed of $\sqrt{u^2+v^2}$ as seen from B. Thus, time for the nearest distance, $T = l\cos\theta/\sqrt{u^2+v^2} = \dfrac{lv}{\sqrt{u^2+v^2}} \times \dfrac{1}{\sqrt{u^2+v^2}}$

thus $T = lv/(u^2+v^2)$ Ans.

EXAMPLE: Air speed of an aircraft is u while wind is blowing radially with a speed $v|\cos\theta|$. Find the time taken by aircraft to complete a circle of radius R parallel to earth's surface. Assume $v << u$ and make suitable approximations.



Solution: Consider the aircraft A at an angular position θ. Here the wind speed perpendicular to the circle is $v\cos\theta$, thus one component of aircraft air speed goes to cancel it and the other component helps in tangential movement. Thus speed along tangent,

$v_\tau = \sqrt{u^2 - v^2\cos^2\theta}$

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Now time required to coven an are angle $d\theta$:

$$dt = \frac{R\,d\theta}{\sqrt{u^2 - v^2\cos^2\theta}} = \frac{R}{u}\,d\theta\left[1 - \frac{v^2}{u^2}\cos^2\theta\right]^{-1/2}$$

Now using binomial approximation, $dt = \frac{R}{u}\left[1 + \frac{1}{2}\frac{v^2}{u^2}\cos^2\theta\right]d\theta$

$$\Rightarrow T = \int dt = \frac{R}{u}\int_0^{2\pi}\left(1 + \frac{v^2}{2u^2}\cos^2\theta\right)d\theta = \frac{\pi R}{2u}\left[4 + \frac{v^2}{u^2}\right] \text{ Ans.}$$

→ Note that fractional change in time is very small. For example for $v/u = 1/3$, the fractional change is $1/36$, so for sufficiently small v/u, we have a very good estimate. For more accuracy we could always use higher order terms in binomial expansion.

→ Try at home: What if there was a constant eastward wind velocity v ($<<u$). Ans: Same result.

River Boat Problem:

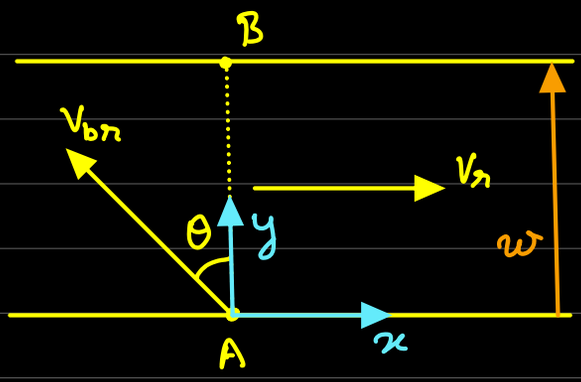
The speed of a boat in still water is v_{bn} while a river flows with a speed v_n as shown we wish to evaluate the following:

(a) Time required for crossing

(b) Drift (x coordinate upon crossing)

(c) Direction of rowing for zero drift (reaching directly opposite point).

(d) Direction of rowing for minimum drift if zero drift is not possible.



Time Required For Crossing:

$$\vec{v}_b = \vec{v}_n + \vec{v}_{bn} \Rightarrow v_{by} = v_{ny} + v_{bny} = 0 + v_{bn}\cos\theta = v_{bn}\cos\theta$$

Now time for crossing $T = S_y/v_{by} = w/v_{bn}\cos\theta$ thus:

$$\boxed{T = w/v_{bn}\cos\theta}$$

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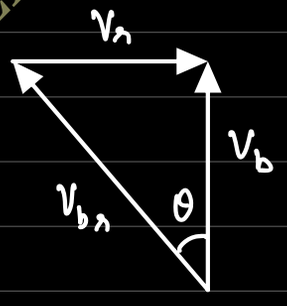
DRIFT: $S_x = V_{bx}\, t = (V_x - V_{by}\sin\theta) \times w/V_{by}\cos\theta$

Thus drift $x_D = w\left[\frac{V_x}{V_{by}}\sec\theta - \tan\theta\right]$

ZERO DRIFT CONDITION: For zero drift, boat should have no velocity along the x direction thus: $V_x - V_{by}\sin\theta = 0 \Rightarrow \sin\theta = V_x/V_{by}$.

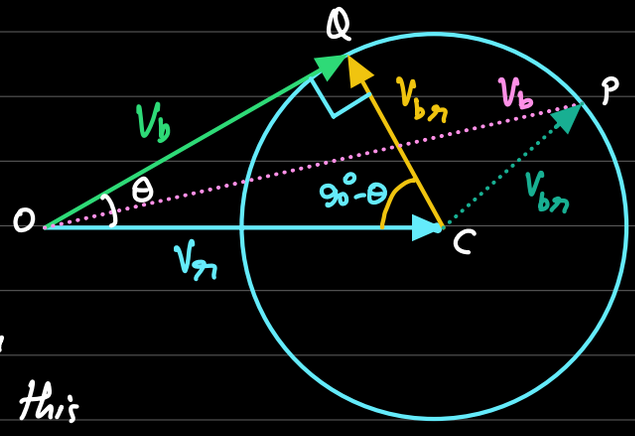
Thus for zero drift $\boxed{\sin\theta = V_x/V_{by}}$

We could also derive this result using triangle law: The resultant of $\vec{V}_x$ and $\vec{V}_{by}$ must be $\perp$ to $\vec{V}_x$. This gives us the required vector diagram as shown in the adjacent figure. From this we can directly see that $\sin\theta = V_x/V_{by}$.



MINIMUM DRIFT CONDITION WHEN ZERO DRIFT IS IMPOSSIBLE:

From the previous section it is clear that zero drift is possible only if $V_x < V_{by}$ $\because |\sin\theta| < 1$. When $V_x > V_{by}$ we wish to minimize the inevitable drift. We could do this using calculus, but there's a more fun geometrical way of visualising this problem. For minimum drift, $\vec{V}_b$ should be inclined at as large angle as possible with $\vec{V}_x$ (or at as small angle as possible with cross line AB). To visualise resultant of $\vec{V}_x$ and $\vec{V}_{by}$ for $\vec{V}_{by}$ oriented at various angles, we draw a vector of length $\vec{V}_x$ and from its head we draw a circle of radius V_{by}. Now suppose we choose V_{by} along OP then the resultant velocity $\vec{V}_b$ is along $\vec{OP}$. Can we make V_b at a larger angle to $\vec{V}_x$? Yes we can see that angle between $\vec{V}_b$ and $\vec{V}_x$ is largest when V_{by} is chosen such that Q is the point of tangency from tail of V_x to the circle.



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Thus, required angle is given by: $\sin\theta = V_{br}/V_r$. Thus for minimum inevitable drift,

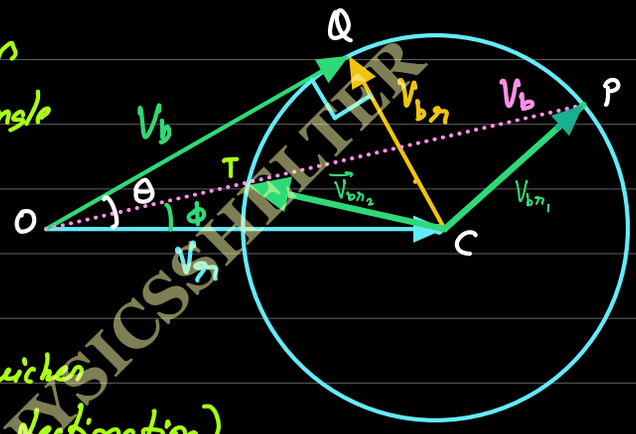
$$\boxed{\sin\theta = V_{br}/V_r}$$

→ RHS here is just the reciprocal of the zero drift case and here $V_{br} < V_r$ while for zero drift, $V_{br} > V_r$.

→ In general to achieve $\vec{V}_b$ at a certain angle to $\vec{V}_r$ which is less than that for minimum drift, there will be two possible directions for choosing $\vec{V}_{br}$.

For instance refer to the adjacent vector diagram. Here to achieve a $\vec{V}_b$ at angle ϕ with $\vec{V}_r$ we could choose $\vec{V}_{br}$ either along $\vec{CT}$ or along $\vec{CP}$.

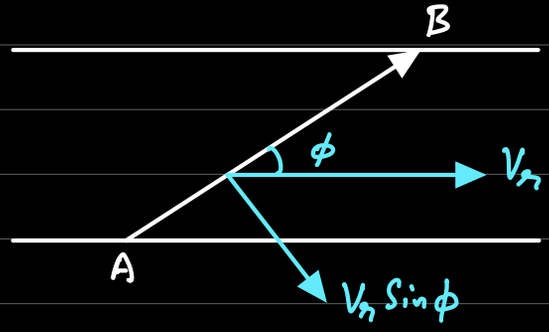
(Though $\vec{CP}$ is obviously better because it leads to a bigger $\vec{V}_b$ and hence quicker crossing time to reach the same final destination).



→ It is impossible to achieve an angle greater than θ between $\vec{V}_b$ and $\vec{V}_r$, so points on opposite bank an angles greater than this θ wrt river flow are inaccessible, no matter what the rowing direction.

Finding Components Of V_{br} To Reach A Particular Point:

Suppose we wish to cross to a point B such that $\vec{V}_b$ is along AB which is at an angle ϕ with river flow. We can resolve river velocity in to components parallel and $\perp$ to the direction $\vec{AB}$, now one component of V_{br} must cancel the component of V_r $\perp$ to direction AB so that net velocity may be along AB the other component can be chosen along $\vec{AB}$ (wise choice) or against $\vec{AB}$ (Foolish choice unless you want some strategic delay).



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Thus component of $\vec{V}_{br}$ along (or against) $\vec{AB}$ becomes $\sqrt{V_{br}^2 - (V_r \sin\phi)^2}$ and accordingly net V_b becomes either $V_r \cos\phi + \sqrt{V_{br}^2 - (V_r \sin\phi)^2}$ or

$V_r \cos\phi - \sqrt{V_{br}^2 - (V_r \sin\phi)^2}$ (foolish choice).

→ The category of problem involving flying aeroplanes is wind or ants moving on conveyor belts are just analogous to the river boat problems with wind and conveyor being analogues of the flowing river and the aircraft and ant being analogues of the boat.

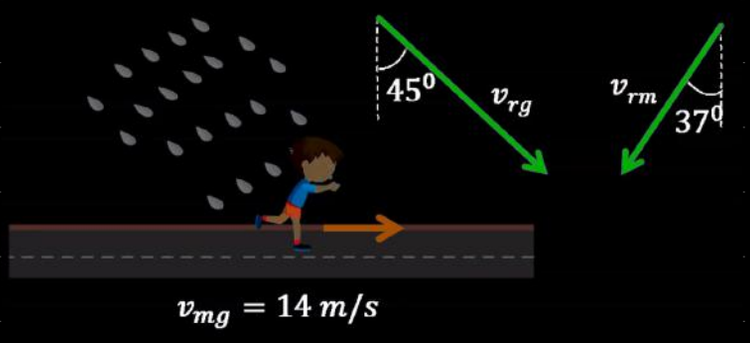
→ By changing frames sometimes problems become analogues of river boat problems, for example from frame of reference of a truck moving with constant velocity, the road looks like a flowing river and people moving on road like boats moving on the flowing river.

RAIN MAN PROBLEM: This is a category of problems based on apparent angle at which rain appears to fall to a moving observer. These type of problems can be solved either by drawing a suitable triangle diagram of vector addition or resolving vectors in to cartesian components.

EXAMPLE: A boy is moving on a horizontal road at a speed of 14 m/s. Rain is falling at an angle 45° inclined in forward direction. To the boy it appears to fall at 37° with vertical coming from the forward direction. What is the actual velocity of rain?

Solution: **METHOD I:** Let velocity of rain be $\vec{V}_r = a\hat{i} - a\hat{j}$ (∵ vector is in fourth quadrant). Now velocity of man, $V_m = 14\hat{i}$.

Now $\vec{V}_{rm} = (a-14)\hat{i} - a\hat{j}$



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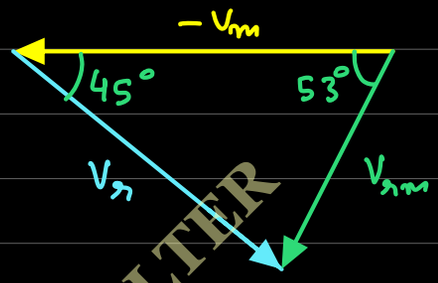
Now this vector is inclined at $(-90-37°)$ with x axis, thus

$V_{rmy}/V_{rm_x} = \tan(-127°) = -\tan(127°) = -\tan(180°-53°) = 4/3$

Thus: $[-a/(a-14)] = 4/3 \Rightarrow a = 8$, thus $\vec{V}_r = 8\hat{i} - 8\hat{j}$ ms^{-1} Ans.

METHOD II (Using Vector Diagram):

Consider the adjacent triangle diagram of vector addition. Applying sine law:



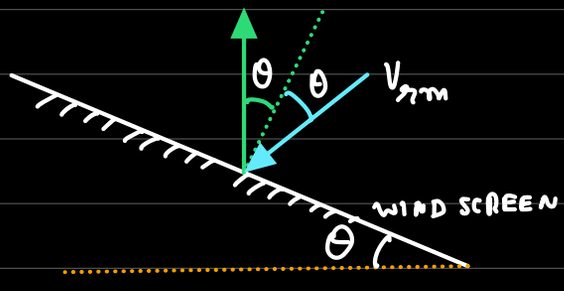
$$\frac{V_r}{\sin 53°} = \frac{V_{rm}}{\sin(180°-(45°+53°))} = \frac{V_{rm}}{\sin 45°}$$

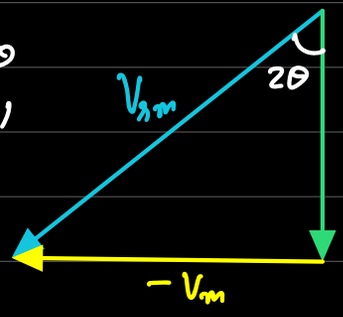
$$\Rightarrow V_r = \frac{14 \times \sin 53°}{\sin 45° \cos 53° + \cos 45 \sin 53°} = \frac{14 \times 4/5}{1/\sqrt{2}\,(3/5+4/5)} = 8\sqrt{2} \text{ ms}^{-1}$$

Thus $V_r = 8\hat{i} - 8\hat{j}$ Ans.

EXAMPLE: Hailstones are falling vertically with a speed u. The windscreen of a car is inclined at an angle θ with horizontal and to the driver, the hailstone appears to bounce vertically after striking the windscreen. Assuming the collisions to be perfectly elastic find the speed of the vehicle.

Solution: Consider the adjacent figure to visualize the bouncing process. Here V_{rm} represents speed of hailstones as observed by the driver. Now we can draw a vector diagram for analysing the situation. From adjacent diagram, clearly we can see that $V_{rm} = u\tan 2\theta$ thus vehicle is moving at a speed equal to $u\tan 2\theta$ Ans.



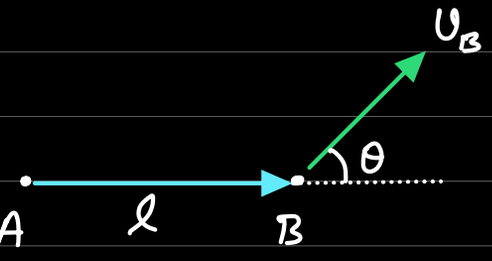


→ θ should be less than 45° if the vehicle is moving forward.

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Velocity Of Separation:

The rate at which distance between two bodies is increasing is called velocity of separation. For instance in the adjacent figure, A is at rest while B is moving with a speed v_B at an angle θ to $\vec{AB}$. The instantaneous separation between them is l.



Then we define the velocity of separation between them as:

$$v_s = dl/dt$$

, if the particles are moving further apart, the velocity of separation is positive and if they are coming closer, the velocity of separation is negative. (Negative of velocity of separation is called velocity of approach).

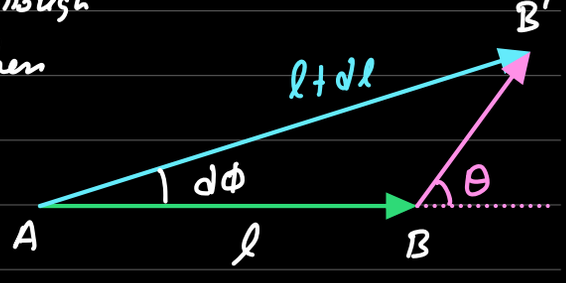
Velocity Of Approach: The rate at which separation between two particles is decreasing is called velocity of approach. Mathematically,

$$v_a = -dl/dt$$

Angular Velocity Of One Particle About The Other:

If the line joining two particles turns through an angle $d\phi$ in a time interval dt, then we define angular velocity as:

$$\omega = d\phi/dt$$

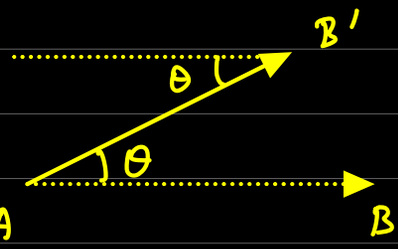


This could be either clockwise or anticlockwise for motion in the plane of the page.

→ Sometimes we hear the words angular velocity of particle A about particle B or angular velocity of particle B about particle A, both are essentially the same thing. If B rotates 30° clockwise wrt its original line joining A, then A also rotates 30° clockwise wrt that line. Yes! "Clockwise" for both, you read that

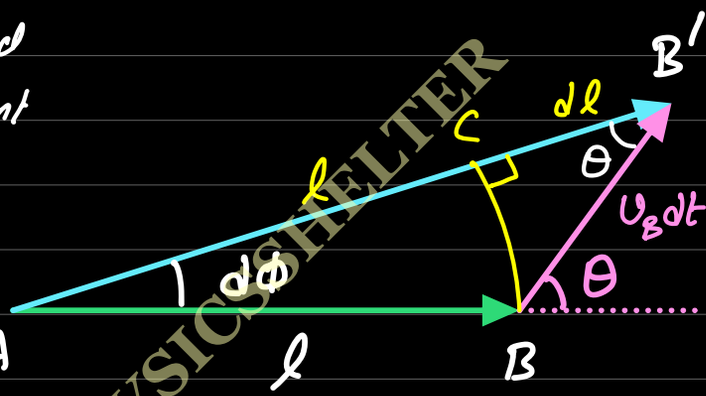
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right. Think carefully. Look at the adjacent figure. To A B has moved through an angle θ clockwise, while for B, A has moved through an angle θ clockwise.



Relation Between Velocity Vectors Of Particles And Velocity Of Separation:

Suppose particle A is at rest and particle B is moving with a speed v_B as shown and reaches a point B' in a time interval dt. To find change in separation we put compass at A and pencil at B to make an arc BC. Now CB' is the change in separation (i.e. dl).



The figure shown is highly exaggerated. Imagine $d\phi$ to be infinitesimal then BCB' approaches a right triangle, and AB' looks nearly ∥ to AB with BB' being its transversal. Thus $\angle CB'B$ is also almost θ.

Now distance $CB' = dl = BB' \cos\theta = v_B\, dt \cos\theta$, thus:

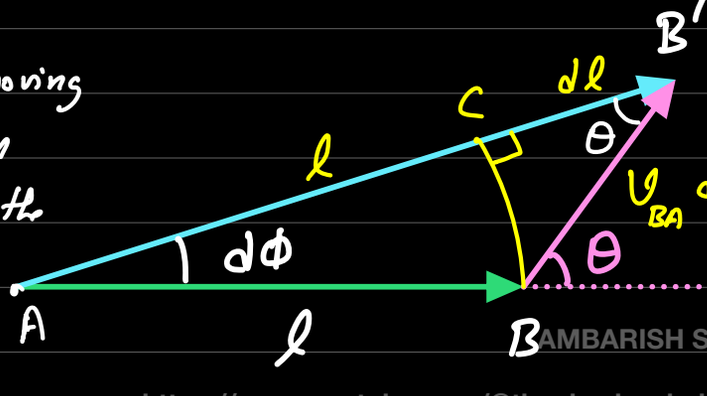
$dl/dt = v_B \cos\theta$, i.e. separation velocity:

$$\boxed{v_s = v_B \cos\theta}$$

This can also be written as $\boxed{v_s = \vec{v_B} \cdot \hat{AB}}$, here $\hat{AB}$ represents unit vector along $\vec{AB}$.

Velocity Of Separation When Both Particles Are Moving:

If both particles A and B are moving we can always convert this problem to the previous case by going in the frame of particle A. (A obviously appears at rest as seen from A itself).

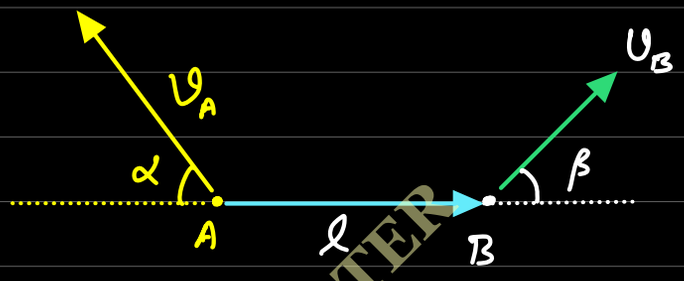


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However in this frame B appears to move with a velocity $\vec{v}_{BA}$ thus separation velocity when both particles are moving is given as

$$\boxed{v_s = \vec{v}_{BA} \cdot \hat{AB} = \vec{v}_{AB} \cdot \hat{BA}}$$

To get a visual feel for this, consider the adjacent figure, Here,



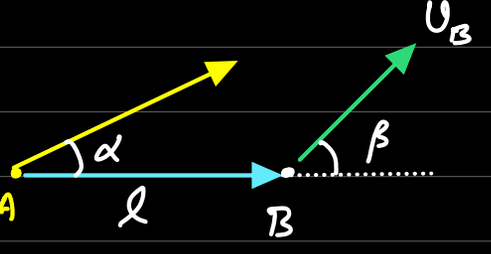
$$v_s = \vec{v}_{BA} \cdot \hat{AB} = (\vec{v}_B - \vec{v}_A) \cdot \hat{AB}$$

$$= \vec{v}_B \cdot \hat{AB} - \vec{v}_A \cdot \hat{AB}$$

$$= v_B \cos\beta + v_A \cos\alpha$$

This is simply a superposition of effects of motion of A and B. If B were moving alone, the particles would be moving apart at a rate $v_B \cos\beta$, while if A were moving alone particles would be moving apart at a rate $v_A \cos\alpha$, thus net velocity of separation can be found by superposing the effects of individual motions of the particles.

As an example consider the adjacent figure. Here due to motion of A alone, the separation will decrease at a rate $v_A \cos\alpha$, while due to motion of B alone, it would increase at a rate $v_B \cos\beta$, thus net separation velocity, $v_s = v_B \cos\beta - v_A \cos\alpha$ and approach velocity is $v_A \cos\alpha - v_B \cos\beta$.



[If separation is decreasing, approach velocity is positive and if separation in increasing, separation velocity is positive].

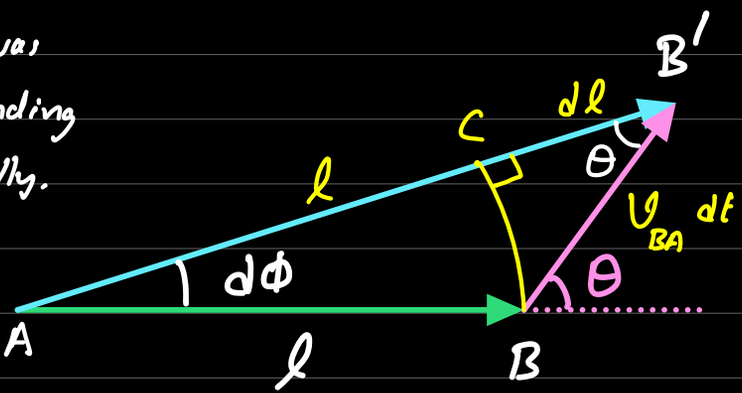
→ Velocity of separation and velocity of approach are misnomers because they are actually just signed scalars by definition.

→ Velocity of separation / approach are simply component of relative velocity of particles along the line joining them.

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Angular Velocity Of One Particle About Another:

Consider the adjacent figure, this was drawn in a previous section for finding the velocity of separation analytically. Now we will use it to find angular velocity of line AB.



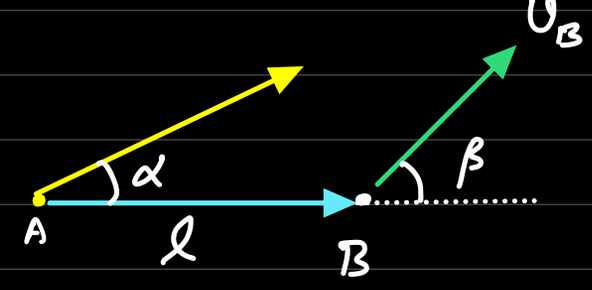
Here the arc length $\overset{\frown}{BC} = l\,d\phi = v_{BA}\,dt\sin\theta$

$\Rightarrow$ Angular velocity, $\omega = \frac{d\phi}{dt} = v_{BA}\sin\theta/l$, thus:

$$\boxed{\omega = \frac{v_{BA}\sin\theta}{l}}$$

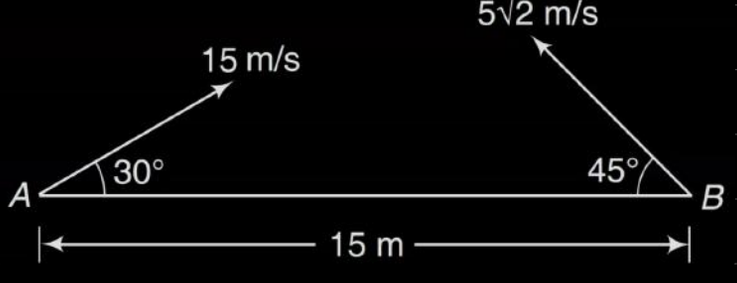
→ Thus angular velocity is simply the component of relative velocity $\perp$ to the line joining the particles divided by their instantaneous distance.

→ Just as with velocity of separation, angular velocity can also be found using superposition. For instance in adjacent figure, due to motion of B alone angular velocity is $v_B\sin\beta/l$ anticlockwise while due to motion of A alone it is $v_A\sin\alpha/l$ clockwise thus net effect,



$$\omega = \frac{v_B\sin\beta}{l} - \frac{v_A\sin\alpha}{l}\circlearrowleft = \frac{v_B\sin\beta - v_A\sin\alpha}{l}\circlearrowleft$$

Example: Two particles A and B are moving as shown in the figure. Find (a) Their velocity of approach, (b) Angular velocity of particles about each other.



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Solution: Using superposition, velocity of approach, $V_a = 15\cos 30° + 5\sqrt{2}\cos 45°$

thus $V_a = \frac{15\sqrt{3}}{2} + 5\ ms^{-1} \approx 18\ ms^{-1}$. Now ω by superposition:

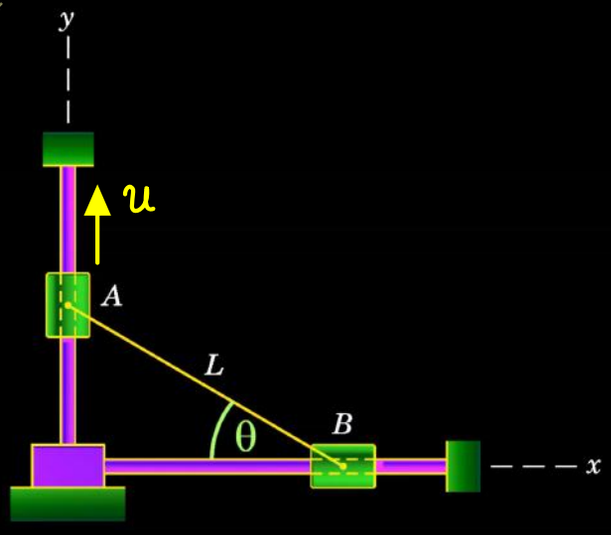
$$(5\sqrt{2}\sin 45°)/15 \circlearrowright + (15\sin 30°/15)\circlearrowleft = \frac{1}{3}\circlearrowright + \frac{1}{2}\circlearrowleft$$

$= \frac{1}{6}\circlearrowleft$ thus angular velocity of particles about each other is $1/6\ rad\,s^{-1}$ clockwise.

Think: If AB were a uniform elastic string what will be velocity of it's mid point (on a general point dividing AB in ratio $\lambda : \mu$).

Solution: If it is a uniform elastic massless string, distances between particles would scale proportionately so we could use section formula to find velocity of any particle. Since all particles of string remain collinear we could also say that angular speed of every particle about every other particle would be same.

EXAMPLE: In the adjacent figure, two collars A and B are connected by an inextensible string. Now A is made to slide upwards with a constant speed u. Find (a) Velocity of B (b) Angular velocity of A about B (c) Acceleration of B (d) Acceleration of mid point of the string.



Solution: Since the distance between A and B is constant (due to string being inextensible), therefore, their velocity of approach must be zero.

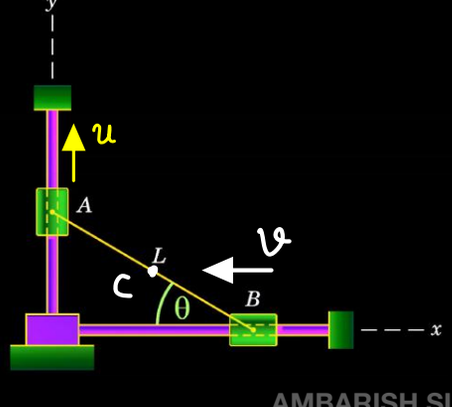
Now let B move with a speed v to the left,

then, $v\cos\theta - u\sin\theta = 0$ (or $v\cos\theta = u\sin\theta$).

Thus, $v = u\tan\theta$ Ans. (Thus $V_B = -u\tan\theta\,\hat{i}$).

(b) Now $\omega = \frac{d\theta}{dt} = \frac{u\cos\theta}{L}\circlearrowleft + \frac{v\sin\theta}{L}\circlearrowleft$

$\Rightarrow \frac{d\theta}{dt} = \frac{u}{L}[\cos\theta + \sin\theta\tan\theta] = \frac{u}{L}\sec\theta$ Ans.



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(c) For finding acceleration we can differentiate the velocity vector, thus:

$$\vec{a}_B = \frac{d\vec{v}_B}{dt} = \frac{d}{dt}[-u\tan\theta\,\hat{i}] = -u\sec^2\theta\frac{d\theta}{dt}\hat{i} = \frac{-u^2}{L}\sec^3\theta\,\hat{i}$$

($\because d\theta/dt = u\sec\theta/l$ as derived for part b), thus $\vec{a}_B = \frac{-u^2}{L}\sec^3\theta\,\hat{i}$ Ans

(d) We can obtain velocity or acceleration of any point on the string using section formula. Thus acceleration of mid point:

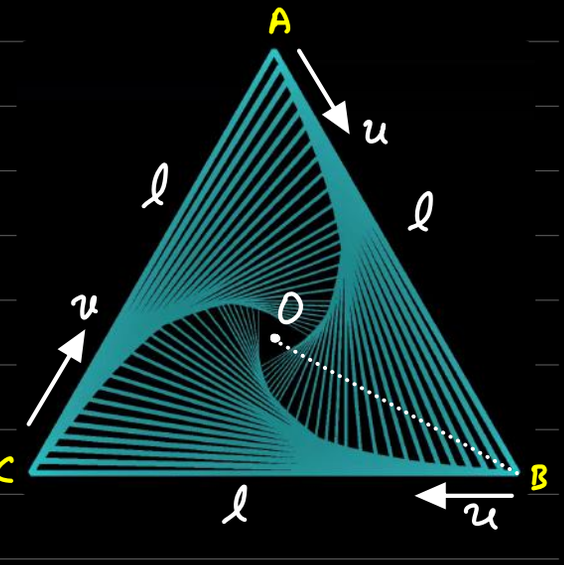
$$a_c = \frac{1}{2}\left[0\hat{j} - \frac{u^2}{L}\sec^3\theta\,\hat{i}\right] = -\frac{u^2}{2L}\sec^3\theta\,\hat{i}$$ Ans.

Think: What is the locus of mid point of the string.

Solution: For mid point, at any general θ, $x = \frac{L}{2}\cos\theta$, $y = L/2\sin\theta$, thus, $x^2 + y^2 = L^2/4$, this is a circle of radius $L/2$ centred at origin.
(For other points on string you can verify that locus will be an ellipse).

→ Whenever two objects are connected to a taut inextensible string, components of their velocities along string are same because their velocity of approach must be zero.

EXAMPLE: Three particles A, B and C move such that A always heads for B, B always heads for C and C always heads for A each moving with a constant speed as shown. Find (a) Time taken for particles to meet (b) length of the path covered



Solution: Due to symmetry, all the three particles will always maintain an equilateral configuration, thus velocities of approach between particles remains constant with time. Now velocity of approach (say of B towards C):

$v_a = u + u\cos 60^\circ = 3u/2$ (This is rate of decrease of separation).

Now time of meeting T = Separation / (rate of decrease of separation)

Thus, $T = l/(3u/2) = 2l/3u$ Ans.

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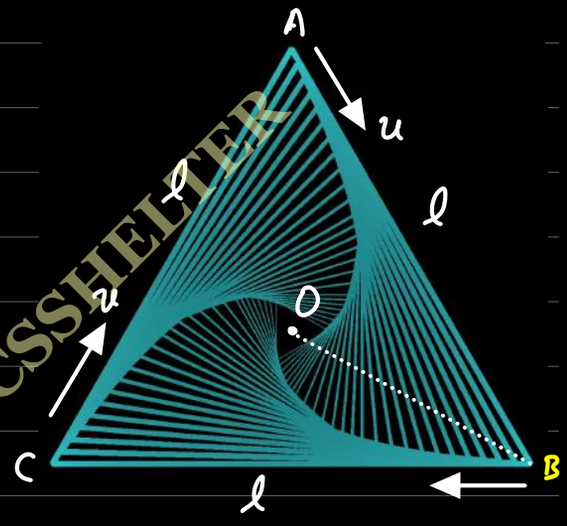
METHOD 2: All the particles meet at centre of the triangle (which is stationary). Now approach velocity towards centre (say of B towards O):

$U_a = u\cos 30^\circ = u\sqrt{3}/2$, Now $T = OB/(u\sqrt{3}/2) = \left(\frac{2}{3}\ell\cos 30^\circ\right)/\left(\frac{u\sqrt{3}}{2}\right)$

$\Rightarrow T = 2\ell/3u$ sec.

(b) Now since speed is constant $\therefore$ length of path: $l = uT = 2\ell/3$ cm.

EXAMPLE: For the situation of the previous problem, (a) What is the initial rate of rotation of velocity vector (b) What is the acceleration at the initial moment (c) What is the radius of curvature of the path at the initial moment?



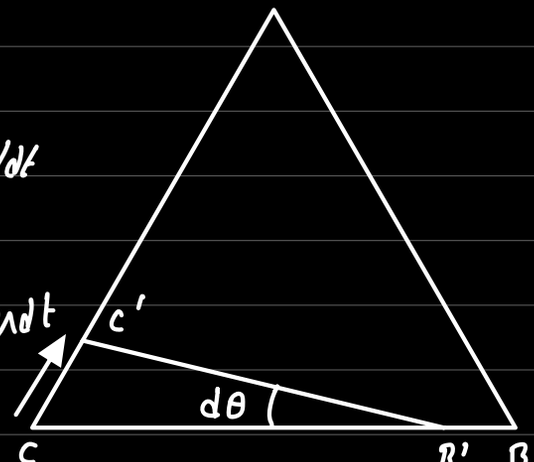
Solution: Since velocity is always directed along the line joining two particles, its rate of rotation must be same as rate of rotation of line joining the two particles. Thus, $d\theta/dt = u\cos 30^\circ/\ell = u\sqrt{3}/2\ell$

Thus $\omega = u\sqrt{3}/2\ell$ (clockwise).

Now recall from lecture of 2D motion that tangential acceleration $a_t = dv/dt$ i.e. d speed/dt which is zero since speed is constant. Also recall that normal acceleration (centripetal acc^n) is $(v\,d\theta/dt)\hat{n}$. Thus for our case, required acc^n $\vec{a} = (u \times u\sqrt{3}/2\ell)\hat{n} = u^2\sqrt{3}/2\ell\,\hat{n}$

($\hat{n}$ is normal of the path towards centre of curvature).

So $\vec{a} = \frac{\sqrt{3}}{2}u^2/\ell\,\hat{n}$ cm.



Now recall that radius of curvature is simply $R = v^2/a_c = u^2/\left(\frac{\sqrt{3}u^2}{2\ell}\right)$

$\Rightarrow R = 2\ell/\sqrt{3}$ cm.

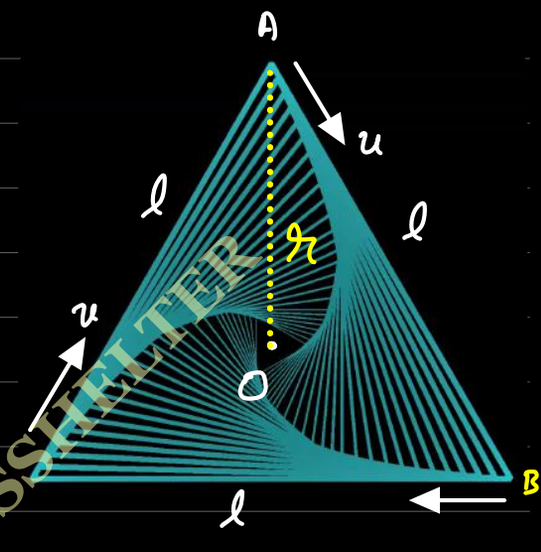
→ Think: What are ω, a and R at a general times?

Solution: We just need to replace ℓ by instantaneous separation. Now

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since velocity of approach is $3u/2$, separation at any general time, $l' = l - 3ut/2$. Now we can just substitute this for l.

EXAMPLE: In the situation of previous problem, let the distance OA at any general time t be r and let θ be the angle through which line OA has rotated, find $r(\theta)$.



Solution: $\omega = \frac{d\theta}{dt} = u\sin 30°/r = u/2r$ ①

$$\frac{dr}{dt} = -u\cos 30° = -u\sqrt{3}/2 \quad ②$$

① / ② $\Rightarrow$

$$\frac{d\theta}{dr} = \frac{1}{\sqrt{3}r} \Rightarrow \frac{dr}{r} = -\sqrt{3}\,d\theta$$

Now @ $\theta = 0$, $r = \frac{2}{3}l\cos 30° = l/\sqrt{3}$, thus:

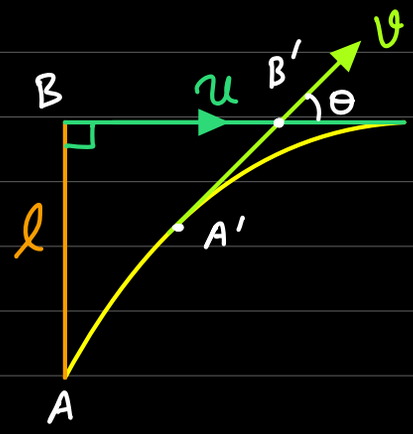
$$\int_{l/\sqrt{3}}^{r} \frac{dr}{r} = -\sqrt{3}\int_0^{\theta} d\theta \Rightarrow \ln[r/(l/\sqrt{3})] = -\sqrt{3}\theta \Rightarrow r = \frac{l}{\sqrt{3}}e^{-\sqrt{3}\theta}$$ Ans.

→ This is the equation of a logarithmic spiral.

EXAMPLE: IRODOV 1.13:

Point A moves uniformly with velocity v so that the vector v is continually "aimed" at point B which in its turn moves rectilinearly and uniformly with velocity $u < v$. At the initial moment of time $v \perp u$ and the points are separated by a distance l. How soon will the points converge?

Solution: Let the velocity vector of A be inclined at an angle θ to the line of motion of B as shown. Now let T be the time of convergence. At this time horizontal coordinate of both particles must be same, thus:



$$uT = \int_0^T v\cos\theta\,dt = v\int_0^T \cos\theta\,dt \quad ①$$

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Also, approach velocity of A, $v_a = v - u\cos\theta$, now since approach velocity is rate of decrease of separation, it's time integral should give total decrease in separation, thus:

$$\int_0^T (v - u\cos\theta)\,dt = vT - u\int_0^T \cos\theta\,dt = l \quad (2)$$

Now substituting $\int_0^T \cos\theta\,dt$ from equation (1), :

$$vT - u \times \frac{u}{v} T = l \implies T = \frac{lv}{v^2 - u^2} \text{ sec.}$$

→ For catching in a finite time, $v > u$.

EXAMPLE: For the situation of previous problem suppose r_0 is instantaneous separation between A and B, find $r(\theta)$.

Solution:

$$\frac{dr}{dt} = -(v - u\cos\theta), \quad \frac{d\theta}{dt} = -u\sin\theta/r, \text{ thus:}$$

$$\frac{dr}{d\theta} = \left(\frac{v - u\cos\theta}{u\sin\theta}\right) r \implies \int_l^{r_0} \frac{dr}{r} = \int_{\pi/2}^{\theta} \left(\frac{v - u\cos\theta}{u\sin\theta}\right) d\theta$$

$$\implies \ln(r_0/l) = \frac{v}{u} \ln[\tan(\theta/2)] - \ln[\sin\theta] \text{ sec.}$$

$$\implies r = l\frac{[\tan(\theta/2)]^{v/u}}{\sin\theta} \text{ sec.}$$

If $v = u$, $r = l\left(\frac{\tan\theta/2}{2\sin\theta/2\cos\theta/2}\right) = \frac{l}{2\cos^2\theta/2} = \frac{l}{1+\cos\theta}$

→ This is polar equation of a parabola. (General equation of conic section : $r = \frac{l}{1 - e\cos\theta}$ where e is eccentricity, if θ were measured from opposite direction, we would see $r = \frac{l}{1-\cos\theta}$)

Thus for A, B appears to move in parabola in this case

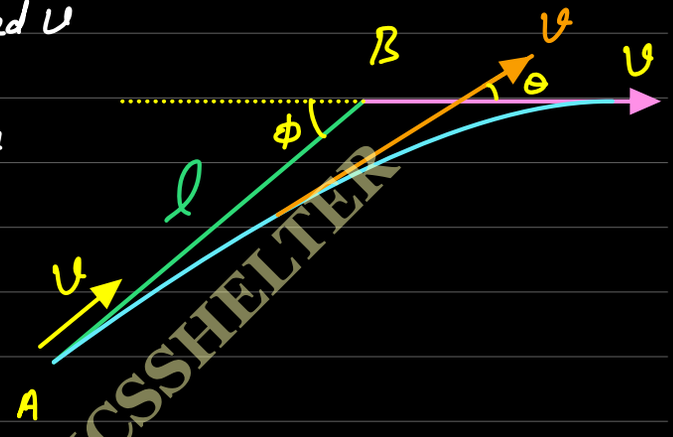
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→ Some more integration formulae: $\int \tan\theta\, d\theta = \ln(\sec\theta) + C$

$\int \cot\theta\, d\theta = \ln(\sin\theta) + C = -\ln(\operatorname{cosec}\theta) + C$, $\int \sec\theta\, d\theta = \ln|\sec\theta + \tan\theta|$

$\int \operatorname{cosec}\theta\, d\theta = \ln|\operatorname{cosec}\theta - \cot\theta| + C$.

EXAMPLE: Suppose A and B had same speed v and at the beginning of the chase, A is located at angle ϕ and distance l wrt b as shown. What would be the final separation between A and B after sufficiently long time.



Solution: Let final separation be l'. Then:

$$\int_0^T (v - v\cos\theta)\, dt = l - l' \quad (1)$$

Finally the particles are moving with almost collinear velocities. Now difference in horizontal coordinates:

$$vT - \left[\int v\cos\theta\, dt - l\cos\phi\right] = l' \quad (2)$$

$(1) - (2) \Rightarrow -l\cos\phi = l - 2l' \Rightarrow l' = (l + l\cos\phi)/2$ Ans.

EXAMPLE: What is the initial acceleration and radius of curvature of trajectory of A in the situation of previous problem.

Solution: Since speed is constant, tangential acceleration is zero. Now

$\omega = v\sin\phi/l$, $a_c = v\omega = v^2\sin\phi/l$ Ans. $R_c = v^2/a_c$

$\Rightarrow$ Radius of curvature, $R = l/\sin\phi$ Ans.

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